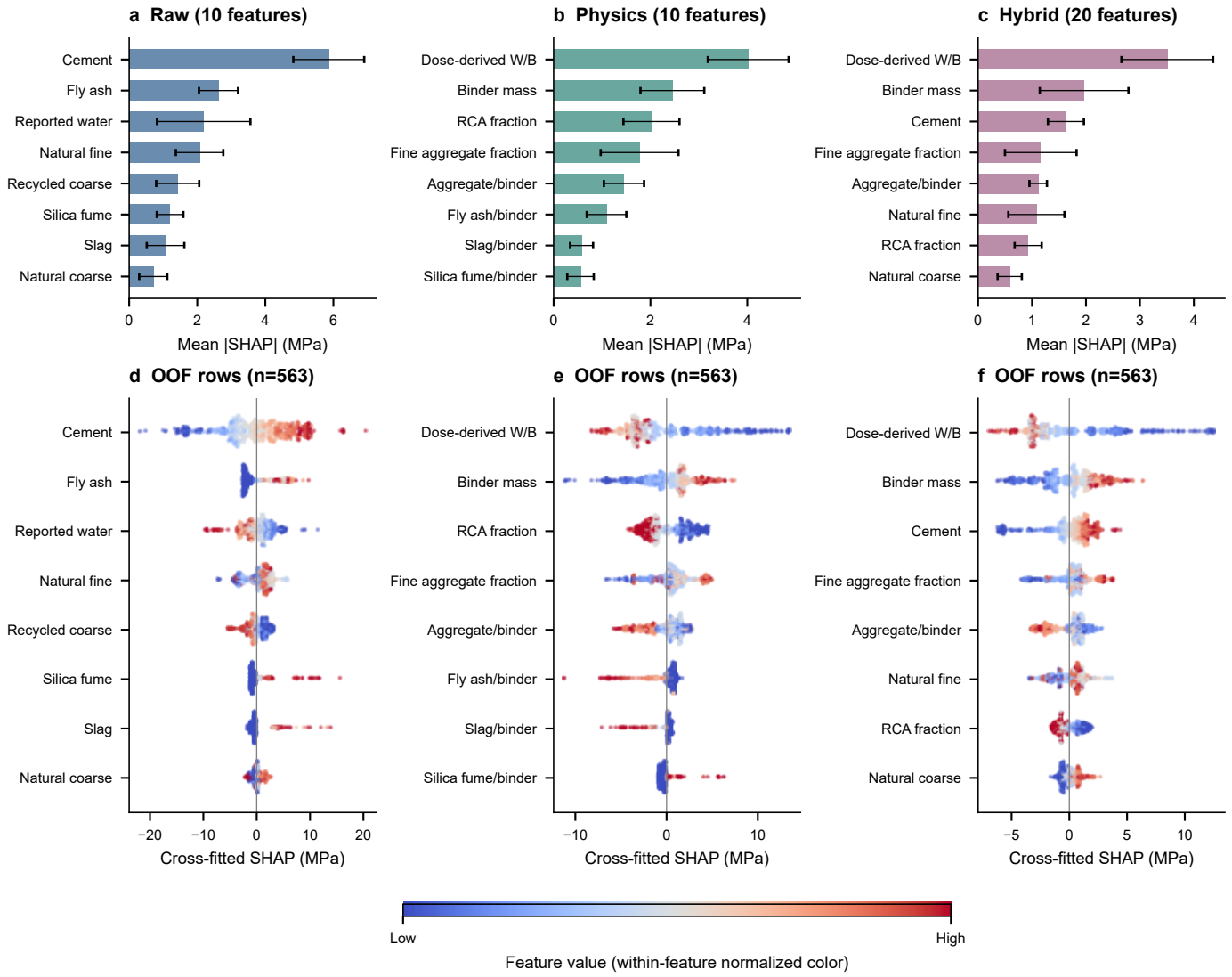

a Raw (10 features)
Cement
Fly ash
Reported water
Natural fine
Recycled coarse
Silica fume
Slag
Natural coarse
0
2
4
6
Mean |SHAP| (MPa)
b Physics (10 features)
Dose-derived W/B
Binder mass
RCA fraction
Fine aggregate fraction
Aggregate/binder
Fly ash/binder
Slag/binder
Silica fume/binder
0
2
4
Mean |SHAP| (MPa)
c Hybrid (20 features)
Dose-derived W/B
Binder mass
Cement
Fine aggregate fraction
Aggregate/binder
Natural fine
RCA fraction
Natural coarse
0
1
2
3
4
Mean |SHAP| (MPa)
d OOF rows (n=563)
Cement
Fly ash
Reported water
Natural fine
Recycled coarse
Silica fume
Slag
Natural coarse
−20
−10
0
10
20
Cross-fitted SHAP (MPa)
e OOF rows (n=563)
Dose-derived W/B
Binder mass
RCA fraction
Fine aggregate fraction
Aggregate/binder
Fly ash/binder
Slag/binder
Silica fume/binder
−10
0
10
Cross-fitted SHAP (MPa)
f OOF rows (n=563)
Dose-derived W/B
Binder mass
Cement
Fine aggregate fraction
Aggregate/binder
Natural fine
RCA fraction
Natural coarse
−5
0
5
10
Cross-fitted SHAP (MPa)
Low
High
Feature value (within-feature normalized color)